

KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5082

5 Credits: 4

Program Core

Source-grounded

Industrial Automation and

□ To understand the basic concepts of different control strategies.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 5082 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5082.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Instrumentation Engineering
Course code	5082
Course title	Industrial Automation and
Semester	5 Credits: 4
Course category	Program Core
Periods per week	4 Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

21 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD**How to use this lesson****First pass**

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- □ To understand the basic concepts of different control strategies.
- □ To describe the use of computers in process Industry
- □ To solve the automation related problems using PLC ladder logic programming
- □ To get the knowledge about various modern control strategies
- □ To conceptualize safety and documentation in process fields

Course outcomes

CO	Official outcome	Study level
CO1	Describe various process control strategies 15 Understanding	See official syllabus
CO2	Solve the engineering problems using PLC 14 Applying	See official syllabus
CO3	Describe Modern Control Strategies 14 Understanding Explain Safety and documentation in process	See official syllabus
CO4	15 Understanding industries	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 3 3
CO3	CO3 2
CO4	CO4 2 3 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Describe various process control strategies	4	As specified
Module 2	engineering problems using PLC	5	As specified
Module 3	Describe Modern Control Strategies	7	As specified
Module 4	Explain Safety and documentation in process industry	5	As specified

MODULE 1

Describe various process control strategies

This module is presented from the official syllabus scope for Industrial Automation and. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Describe various process control strategies
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ Explain different types of process variables 3 Understanding Compare Feedback and Feed forward control

Explain different types of process variables 3 Understanding Compare Feedback and Feed forward control is an official topic in Module 1 of Industrial Automation and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain different types of process variables 3 Understanding Compare Feedback and Feed forward control using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain different types of process variables 3 understanding compare feedback and feed forward control should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain different types of process variables 3 Understanding Compare Feedback and Feed forward control means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Explain different types of process variables 3 Understanding Compare Feedback and Feed forward control
 പരസ്പരം വിശദീകരിക്കുക definition/meaning വിശദീകരിക്കുക. പരസ്പരം വിശദീകരിക്കുക
 വിശദീകരിക്കുക, steps, application വിശദീകരിക്കുക വിശദീകരിക്കുക. Technical terms
 English- വിശദീകരിക്കുക വിശദീകരിക്കുക.

▼ 4 Understanding system Explain Cascade control system, Ratio control

4 Understanding system Explain Cascade control system, Ratio control is an official topic in Module 1 of Industrial Automation and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding system Explain Cascade control system, Ratio control using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding system explain cascade control system, ratio control should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 4 Understanding system Explain Cascade control system, Ratio control means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-4 Understanding system Explain Cascade control system, Ratio control ഏകാക്ഷരം ഏകാക്ഷരം definition/meaning ഏകാക്ഷരം. ഏകാക്ഷരം ഏകാക്ഷരം ഏകാക്ഷരം, steps, application ഏകാക്ഷരം ഏകാക്ഷരം ഏകാക്ഷരം. Technical terms English-ഐ ഏകാക്ഷരം ഏകാക്ഷരം.

▼ 4 Understanding system and Adaptive control system Explain Inferential control system, Split

4 Understanding system and Adaptive control system Explain Inferential control system, Split is an official topic in Module 1 of Industrial Automation and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding system and Adaptive control system Explain Inferential control system, Split using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding system and adaptive control system explain inferential control system, split should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 4 Understanding system and Adaptive control system Explain Inferential control system, Split means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-4 Understanding system and Adaptive control system Explain Inferential control system, Split definition/meaning. steps, application. Technical terms English- Malayalam.

▼ **Range Control system, Override control 4 Understanding system and Auctioneering control system Basic Process control strategies -Single variable - Independent variable - Interactive variable - compound variable - multivariable control (with examples) - Feedback control system - Feed forward control system - Comparison - Cascade control system - Ratio control system - Adaptive control system -Inferential control system - Split Range Control system - Override control system-Auctioneering control system. Explain the use of Computers in Process Control and solve the**

Range Control system, Override control 4 Understanding system and Auctioneering control system Basic Process control strategies -Single variable - Independent variable - Interactive variable - compound variable - multivariable control (with examples) - Feedback control system - Feed forward control system - Comparison - Cascade control system - Ratio control system - Adaptive control system -Inferential control system - Split Range Control system - Override control system-Auctioneering control system. Explain the use of Computers in Process Control and solve the is an official topic in Module 1 of Industrial Automation and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Range Control system, Override control 4 Understanding system and Auctioneering control system Basic Process control strategies - Single variable - Independent variable - Interactive variable - compound variable - multivariable control (with examples) - Feedback control system -

Feed forward control system - Comparison - Cascade control system - Ratio control system - Adaptive control system - Inferential control system - Split Range Control system - Override control system - Auctioneering control system. Explain the use of Computers in Process Control and solve the using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, range control system, override control 4 understanding system and auctioneering control system basic process control strategies - single variable - independent variable - interactive variable - compound variable - multivariable control (with examples) - feedback control system - feed forward control system - comparison - cascade control system - ratio control system - adaptive control system - inferential control system - split range control system - override control system - auctioneering control system. explain the use of computers in process control and solve the should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Range Control system, Override control 4 Understanding system and Auctioneering control system Basic Process control strategies - Single variable - Independent variable - Interactive variable - compound variable - multivariable control (with examples) - Feedback control system - Feed forward control system - Comparison - Cascade control system - Ratio control system - Adaptive control system - Inferential control system - Split Range Control system - Override control system - Auctioneering control system. Explain the use of Computers in Process Control and solve the means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-റേഞ്ച് Range Control system, Override control 4 Understanding system and Auctioneering control system Basic Process control strategies -Single variable - Independent variable - Interactive variable - compound variable - multivariable control (with examples) - Feedback control system - Feed forward control system - Comparison - Cascade control system - Ratio control system - Adaptive control system - Inferential control system - Split Range Control system - Override control system-Auctioneering control system. Explain the use of Computers in Process Control and solve the ഏകദേശം definition/meaning ഏകദേശം. ഏകദേശം ഏകദേശം ഏകദേശം, steps, application ഏകദേശം ഏകദേശം. Technical terms English-റേഞ്ച് ഏകദേശം.

► Official syllabus detail and terminology (expand in digital version)

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

engineering problems using PLC

This module is presented from the official syllabus scope for Industrial Automation and. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: engineering problems using PLC
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

▼ System, Supervisory control and Direct 4 Understanding Digital Control

System, Supervisory control and Direct 4 Understanding Digital Control is an official topic in Module 2 of Industrial Automation and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify System, Supervisory control and Direct 4 Understanding Digital Control using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, system, supervisory control and direct 4 understanding digital control should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What System, Supervisory control and Direct 4 Understanding Digital Control means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-System, Supervisory control and Direct 4 Understanding Digital Control നോക്കൂ definition/meaning നോക്കൂ. നോക്കൂ നോക്കൂ, steps, application നോക്കൂ നോക്കൂ. Technical terms English-നോക്കൂ നോക്കൂ.

▼ Describe Distributed control system 2 Understanding Explain Programmable Logic Controller and

Describe Distributed control system 2 Understanding Explain Programmable Logic Controller and is an official topic in Module 2 of Industrial Automation and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe Distributed control system 2 Understanding Explain Programmable Logic Controller and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe distributed control system 2 understanding explain programmable logic controller and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe Distributed control system 2 Understanding Explain Programmable Logic Controller and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-Describe Distributed control system 2 Understanding Explain Programmable Logic Controller and definition/meaning . steps, application . Technical terms English- .

▼ 3 Understanding its application

3 Understanding its application is an official topic in Module 2 of Industrial Automation and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding its application using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding its application should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding its application means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-3 Understanding its application
 പദപരിചയം definition/meaning. പദപരിചയം, steps, application പദപരിചയം പദപരിചയം. Technical terms English-ൽ പദപരിചയം പദപരിചയം.

▼ Solve engineering problems using PLC 2 Applying Explain Supervisory Control And Data

Solve engineering problems using PLC 2 Applying Explain Supervisory Control And Data is an official topic in Module 2 of Industrial Automation and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Solve engineering problems using PLC 2 Applying Explain Supervisory Control And Data using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, solve engineering problems using plc 2 applying explain supervisory control and data should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Solve engineering problems using PLC 2 Applying Explain Supervisory Control And Data means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Solve engineering problems using PLC 2 Applying Explain Supervisory Control And Data definition/meaning . . . , steps, application Technical terms English-

▼ **3 Understanding Acquisition System Computers in process control-Data Loggers with block diagram - Data Acquisition System with Block Diagram - Supervisory control with block diagram - Direct Digital Control with block diagram - Comparison - DCS with its architecture - general features - advantages - PLC - basic concept - Block diagram of PLC operation - input Module, Output Module - PLC scan time - List the programming methods - ladder diagram programming - simple program - SCADA with block diagram - general features - system components**

3 Understanding Acquisition System Computers in process control-Data Loggers with block diagram - Data Acquisition System with Block Diagram - Supervisory control with block diagram - Direct Digital Control with block diagram - Comparison - DCS with its architecture - general features - advantages - PLC - basic concept - Block diagram of PLC operation - input Module, Output Module - PLC scan time - List the programming methods - ladder diagram programming - simple program - SCADA with block diagram - general features - system components is an official topic in Module 2 of Industrial Automation and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding Acquisition System Computers in process control-Data Loggers with block diagram - Data Acquisition System with Block Diagram - Supervisory control with block diagram - Direct Digital Control with block diagram - Comparison - DCS with its architecture - general features - advantages - PLC - basic concept - Block diagram of PLC

operation - input Module, Output Module - PLC scan time - List the programming methods - ladder diagram programming - simple program - SCADA with block diagram - general features - system components using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding acquisition system computers in process control-data loggers with block diagram - data acquisition system with block diagram - supervisory control with block diagram - direct digital control with block diagram - comparison - dcs with its architecture - general features - advantages - plc - basic concept - block diagram of plc operation - input module, output module - plc scan time - list the programming methods - ladder diagram programming - simple program - scada with block diagram - general features - system components should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding Acquisition System Computers in process control-Data Loggers with block diagram - Data Acquisition System with Block Diagram - Supervisory control with block diagram - Direct Digital Control with block diagram - Comparison - DCS with its architecture - general features - advantages - PLC - basic concept - Block diagram of PLC operation - input Module, Output Module - PLC scan time - List the programming methods - ladder diagram programming - simple program - SCADA with block diagram - general features - system components means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-3 Understanding Acquisition System Computers in process control-Data Loggers with block diagram - Data Acquisition System with Block Diagram - Supervisory control with block diagram - Direct Digital Control with block diagram - Comparison - DCS with its architecture - general features - advantages - PLC - basic concept - Block diagram of PLC operation - input Module, Output Module - PLC scan time - List the programming methods - ladder diagram programming - simple program - SCADA with block diagram - general features - system components
 താഴെ പറയുന്നവയുടെ നിർവ്വചനം/അർത്ഥം എഴുതുക.
 താഴെ പറയുന്നവയുടെ ഘട്ടം, പ്രയോഗം എഴുതുക.
 Technical terms English-ൽ എഴുതുക.

► Official syllabus detail and terminology (expand in digital version)

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Describe Modern Control Strategies

This module is presented from the official syllabus scope for Industrial Automation and. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Describe Modern Control Strategies
- Official duration: As specified
- Official topic entries captured: 7
- The full source excerpt is preserved below for coverage checking.

▼ Explain Virtual instrumentation

Explain Virtual instrumentation is an official topic in Module 3 of Industrial Automation and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Virtual instrumentation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain virtual instrumentation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain Virtual instrumentation means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Explain Virtual instrumentation
 വിശ്വകോശം വിവരിച്ചിട്ടുള്ള definition/meaning വിശദീകരിക്കുക. വിശദീകരിക്കുക
 വിശദീകരിക്കുക, steps, application വിശദീകരിക്കുക വിശദീകരിക്കുക. Technical terms
 English- വിശദീകരിക്കുക വിശദീകരിക്കുക.

▼ Understand LabVIEW and its applications. 2 Understanding Understand the concept of Artificial Neural

Understand LabVIEW and its applications. 2 Understanding Understand the concept of Artificial Neural is an official topic in Module 3 of Industrial Automation and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand LabVIEW and its applications. 2 Understanding Understand the concept of Artificial Neural using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand labview and its applications. 2 understanding understand the concept of artificial neural should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understand LabVIEW and its applications. 2 Understanding Understand the concept of Artificial Neural means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Understand LabVIEW and its applications. 2 Understanding Understand the concept of Artificial Neural definition/meaning . steps, application . Technical terms English- .

▼ 3 Understanding Network Compare the various Learning methods used

3 Understanding Network Compare the various Learning methods used is an official topic in Module 3 of Industrial Automation and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding Network Compare the various Learning methods used using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding network compare the various learning methods used should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 3 Understanding Network Compare the various Learning methods used means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 3 Understanding Network Compare the various Learning methods used definition/meaning. steps, application. Technical terms English- .

▼ 2 Understanding in ANN

2 Understanding in ANN is an official topic in Module 3 of Industrial Automation and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding in ANN using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding in ann should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding in ANN means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 2 Understanding in ANN ഫലപ്രസാദം
 ഫലപ്രസാദം definition/meaning ഫലപ്രസാദം. ഫലപ്രസാദം ഫലപ്രസാദം, steps,
 application ഫലപ്രസാദം ഫലപ്രസാദം ഫലപ്രസാദം. Technical terms English- ഫലപ്രസാദം
 ഫലപ്രസാദം.

▼ Understand the concept of Fuzzy logic 1 Understanding Explain the Block Diagram of fuzzy

Understand the concept of Fuzzy logic 1 Understanding Explain the Block Diagram of fuzzy is an official topic in Module 3 of Industrial Automation and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand the concept of Fuzzy logic 1 Understanding Explain the Block Diagram of fuzzy using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand the concept of fuzzy logic 1 understanding explain the block diagram of fuzzy should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Understand the concept of Fuzzy logic 1 Understanding Explain the Block Diagram of fuzzy means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Understand the concept of Fuzzy logic 1 Understanding Explain the Block Diagram of fuzzy definition/meaning . . . , steps, application Technical terms English-

▼ 2 Understanding controller Understand Intelligent Control and its

2 Understanding controller Understand Intelligent Control and its is an official topic in Module 3 of Industrial Automation and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding controller Understand Intelligent Control and its using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding controller understand intelligent control and its should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding controller Understand Intelligent Control and its means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 2 Understanding controller Understand Intelligent Control and its definition/meaning. steps, application. Technical terms English- Malayalam.

▼ 2 Understanding Features Modern Control Strategies: Virtual Instrumentation - compare Virtual instrument and traditional Instrument - Hardware and software - Test control and design - LabVIEW - Text based programming and graphical programming - Advantages of LabVIEW - ANN - Biological Neuron - Artificial Neuron Model - Types of activation functions - concepts of Learning methods (supervised, Unsupervised and reinforcement) -Fuzzy Logic - Concept - Classical and fuzzy sets- Properties - Membership function - example - Fuzzy Logic System - Block Diagram - Artificial Intelligence - Intelligent Control - Features

2 Understanding Features Modern Control Strategies: Virtual Instrumentation - compare Virtual instrument and traditional Instrument - Hardware and software - Test control and design - LabVIEW - Text based programming and graphical programming - Advantages of LabVIEW - ANN - Biological Neuron - Artificial Neuron Model - Types of activation functions - concepts of Learning methods (supervised, Unsupervised and reinforcement) -Fuzzy Logic - Concept - Classical and fuzzy sets- Properties - Membership function - example - Fuzzy Logic System - Block Diagram - Artificial Intelligence - Intelligent Control - Features is an official topic in Module 3 of Industrial Automation and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding Features Modern Control Strategies: Virtual Instrumentation - compare Virtual instrument and traditional

Instrument - Hardware and software - Test control and design - LabVIEW - Text based programming and graphical programming - Advantages of LabVIEW - ANN - Biological Neuron - Artificial Neuron Model - Types of activation functions - concepts of Learning methods (supervised, Unsupervised and reinforcement) -Fuzzy Logic - Concept - Classical and fuzzy sets- Properties - Membership function – example - Fuzzy Logic System - Block Diagram - Artificial Intelligence - Intelligent Control – Features using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding features modern control strategies: virtual instrumentation - compare virtual instrument and traditional instrument - hardware and software - test control and design - labview - text based programming and graphical programming - advantages of labview – ann - biological neuron - artificial neuron model - types of activation functions - concepts of learning methods (supervised, unsupervised and reinforcement) - fuzzy logic - concept - classical and fuzzy sets- properties - membership function – example - fuzzy logic system - block diagram - artificial intelligence - intelligent control – features should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding Features Modern Control Strategies: Virtual Instrumentation - compare Virtual instrument and traditional Instrument - Hardware and software - Test control and design - LabVIEW - Text based programming and graphical programming - Advantages of LabVIEW – ANN - Biological Neuron - Artificial Neuron Model - Types of activation functions - concepts of Learning methods (supervised, Unsupervised and reinforcement) -Fuzzy Logic - Concept - Classical and

Aspect	What to prepare
	fuzzy sets- Properties - Membership function - example - Fuzzy Logic System - Block Diagram - Artificial Intelligence - Intelligent Control - Features means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 2 Understanding Features Modern Control Strategies: Virtual Instrumentation - compare Virtual instrument and traditional Instrument - Hardware and software - Test control and design - LabVIEW - Text based programming and graphical programming - Advantages of LabVIEW - ANN - Biological Neuron - Artificial Neuron Model - Types of activation functions - concepts of Learning methods (supervised, Unsupervised and reinforcement) -Fuzzy Logic - Concept - Classical and fuzzy sets- Properties - Membership function - example - Fuzzy Logic System - Block Diagram - Artificial Intelligence - Intelligent Control - Features
 ഏതുതരം ഏതുതരം definition/meaning ഏതുതരം. ഏതുതരം ഏതുതരം
 ഏതുതരം, steps, application ഏതുതരം ഏതുതരം. Technical terms
 English-ൽ ഏതുതരം ഏതുതരം.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Explain Safety and documentation in process industry

This module is presented from the official syllabus scope for Industrial Automation and. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain Safety and documentation in process industry
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

▼ selection of equipments based on hazardous 3 Understanding area Explain the concept of Instrument protection Understanding

selection of equipments based on hazardous 3 Understanding area Explain the concept of Instrument protection Understanding is an official topic in Module 4 of Industrial Automation and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify selection of equipments based on hazardous 3 Understanding area Explain the concept of Instrument protection Understanding using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, selection of equipments based on hazardous 3 understanding area explain the concept of instrument protection understanding should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

▼ 3 and types

3 and types is an official topic in Module 4 of Industrial Automation and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 and types using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 and types should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 and types means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-പാഠ്യ 3 and types പദപരിചയം പദപരിചയം definition/meaning പദപരിചയം പദപരിചയം. പദപരിചയം പദപരിചയം പദപരിചയം, steps, application പദപരിചയം പദപരിചയം പദപരിചയം. Technical terms English-പദ പദപരിചയം പദപരിചയം.

▼ Describe Fire and Gas system

Describe Fire and Gas system is an official topic in Module 4 of Industrial Automation and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe Fire and Gas system using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe fire and gas system should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe Fire and Gas system means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-എന്നു വിവരിക്കുക ഫയറും ഗ്യാസ് സിസ്റ്റം

എന്നതിന്റെ definition/meaning എഴുതുക. എന്നതിന്റെ steps, application എന്നിവ എഴുതുക. Technical terms English-ൽ എഴുതുക.

▼ Explain safety instrumentation system 3 Understanding Explain engineering documents used in

Explain safety instrumentation system 3 Understanding Explain engineering documents used in is an official topic in Module 4 of Industrial Automation and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain safety instrumentation system 3 Understanding Explain engineering documents used in using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain safety instrumentation system 3 understanding explain engineering documents used in should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain safety instrumentation system 3 Understanding Explain engineering documents used in means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Explain safety instrumentation system 3 Understanding Explain engineering documents used in definition/meaning . , steps, application . Technical terms English- .

▼ 3 Understanding Instrumentation Safety and documentation in process industry: Hazardous area classification - basic concept - classification - selection of equipment based on hazardous area - Instrument protection - Concept - types - Fire and Gas (F & G) system - concept and block diagram (types of fire and gas detectors) - Safety Instrumentation System (SIS) - concept and block diagram - different layers - Instrumentation documents - Process Flow Diagrams (PFDs), P&I Diagram -Instrument loop diagrams - Graphical element - line symbol - Instrument identification from P&I Diagram - Examples - Heat exchanger and Distillation column Text / Reference: T/R Book title/Author R1 Curtis .D.Johnson, “Process control instrumentation technology”, PHI R2 George stephanopoulus, “Chemical Process Control”, PHI R3 Krishnakanth, “Computer based Industrial Control”, PHI R4 John webb, “Programmable Logic Controllers”, Macmillan Publishing R5 Liptak,” Handbook of Process Control” R6 Jovitha Jerome,”Virtual Instrumentation using LabVIEW”, PHI Rajasekaran. S. Vijayalakshmi Pai. G.A. “Neural Networks, Fuzzy Logic and R7 Genetic Algorithms”, Prentice Hall of India Private Limited, 2003 Timothy J.Ross, “Fuzzy logic with Engineering Applications”, McGraw Hill, R8 1995 Alireza bahadori, “Hazardous area classification in petroleum and chemical plants R9 a guide to mitigating risk” Online Resources: Sl.No Website Link 1 <https://www.slideshare.net/debuddit/hazardous-area-classification-56437172> 2 <https://youtu.be/IBJnU1MJAts>

3 Understanding Instrumentation Safety and documentation in process industry: Hazardous area classification - basic concept - classification - selection of equipment based on hazardous area - Instrument protection - Concept - types - Fire and Gas (F & G) system - concept and block diagram (types of fire and gas detectors) - Safety Instrumentation System (SIS) - concept and block diagram - different layers - Instrumentation documents - Process Flow Diagrams (PFDs), P&I Diagram -Instrument loop diagrams - Graphical element - line symbol - Instrument identification from P&I Diagram - Examples - Heat exchanger and Distillation column Text / Reference: T/R Book title/Author R1 Curtis .D.Johnson, “Process control

instrumentation technology”, PHI R2 George stephanopoulus, “Chemical Process Control”, PHI R3 Krishnakanth, “Computer based Industrial Control”, PHI R4 John webb, “Programmable Logic Controllers”, Macmillan Publishing R5 Liptak,” Handbook of Process Control” R6 Jovitha Jerome,”Virtual Instrumentation using LabVIEW”, PHI Rajasekaran. S. Vijayalakshmi Pai. G.A. “Neural Networks, Fuzzy Logic and R7 Genetic Algorithms”, Prentice Hall of India Private Limited, 2003 Timothy J.Ross, “Fuzzy logic with Engineering Applications”, McGraw Hill, R8 1995 Alireza bahadori, “Hazardous area classification in petroleum and chemical plants R9 a guide to mitigating risk” Online Resources: Sl.No Website Link 1 <https://www.slideshare.net/debuddit/hazardous-area-classification-56437172> 2 <https://youtu.be/IBJnU1MJAts> is an official topic in Module 4 of Industrial Automation and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding Instrumentation Safety and documentation in process industry: Hazardous area classification - basic concept - classification - selection of equipment based on hazardous area - Instrument protection - Concept - types - Fire and Gas (F & G) system - concept and block diagram (types of fire and gas detectors) - Safety Instrumentation System (SIS) - concept and block diagram - different layers - Instrumentation documents - Process Flow Diagrams (PFDs), P&I Diagram - Instrument loop diagrams - Graphical element - line symbol - Instrument identification from P&I Diagram - Examples - Heat exchanger and Distillation column Text / Reference: T/R Book title/Author R1 Curtis .D.Johnson, “Process control instrumentation technology”, PHI R2 George

stephanopoulos, "Chemical Process Control", PHI R3 Krishnakanth, "Computer based Industrial Control", PHI R4 John webb, "Programmable Logic Controllers", Macmillan Publishing R5 Liptak, "Handbook of Process Control" R6 Jovitha Jerome, "Virtual Instrumentation using LabVIEW", PHI Rajasekaran. S. Vijayalakshmi Pai. G.A. "Neural Networks, Fuzzy Logic and R7 Genetic Algorithms", Prentice Hall of India Private Limited, 2003 Timothy J. Ross, "Fuzzy logic with Engineering Applications", McGraw Hill, R8 1995 Alireza bahadori, "Hazardous area classification in petroleum and chemical plants R9 a guide to mitigating risk" Online Resources: Sl.No Website Link 1 <https://www.slideshare.net/debuddit/hazardous-area-classification-56437172> 2 <https://youtu.be/IBJnU1MJAts> using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding instrumentation safety and documentation in process industry: hazardous area classification - basic concept - classification - selection of equipment based on hazardous area - instrument protection - concept - types - fire and gas (f & g) system - concept and block diagram (types of fire and gas detectors) - safety instrumentation system (sis) - concept and block diagram - different layers - instrumentation documents - process flow diagrams (pfd), p&i diagram - instrument loop diagrams - graphical element - line symbol - instrument identification from p&i diagram - examples - heat exchanger and distillation column text / reference: t/r book title/author r1 curtis .d.johnson, "process control instrumentation technology", phi r2 george stephanopoulos, "chemical process control", phi r3 krishnakanth, "computer based industrial control", phi r4 john webb, "programmable logic controllers", macmillan publishing r5 liptak, "handbook of process control" r6 jovitha jerome, "virtual instrumentation using labview", phi rajasekaran. s. vijayalakshmi pai. g.a. "neural networks, fuzzy logic and r7 genetic algorithms", prentice hall of india private limited, 2003 timothy j.ross, "fuzzy logic with engineering

applications", mcgraw hill, r8 1995 alireza bahadori, "hazardous area classification in petroleum and chemical plants r9 a guide to mitigating risk"

online resources: sl.no website link 1

<https://www.slideshare.net/debuddit/hazardous-area-classification-56437172> 2

<https://youtu.be/ibjnu1mjats> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding Instrumentation Safety and documentation in process industry: Hazardous area classification - basic concept - classification - selection of equipment based on hazardous area - Instrument protection - Concept - types - Fire and Gas (F & G) system - concept and block diagram (types of fire and gas detectors) - Safety Instrumentation System (SIS) - concept and block diagram - different layers - Instrumentation documents - Process Flow Diagrams (PFDs), P&I Diagram -Instrument loop diagrams - Graphical element - line symbol - Instrument identification from P&I Diagram - Examples - Heat exchanger and Distillation column Text / Reference: T/R Book title/Author R1 Curtis .D.Johnson, "Process control instrumentation technology", PHI R2 George stephanopoulos, "Chemical Process Control", PHI R3 Krishnakanth, "Computer based Industrial Control", PHI R4 John webb, "Programmable Logic Controllers", Macmillan Publishing R5 Liptak, "Handbook of Process Control" R6 Jovitha Jerome, "Virtual Instrumentation using LabVIEW", PHI Rajasekaran. S. Vijayalakshmi Pai. G.A. "Neural Networks, Fuzzy Logic and R7 Genetic Algorithms", Prentice Hall of India Private Limited, 2003 Timothy J.Ross, "Fuzzy logic with Engineering Applications", McGraw Hill, R8 1995 Alireza bahadori, "Hazardous area classification in petroleum and chemical plants R9 a guide to mitigating risk" Online Resources: Sl.No Website Link 1 https://www.slideshare.net/debuddit/hazardous-area-classification-56437172 2 https://youtu.be/IBJnU1MJAts means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-3 Understanding Instrumentation Safety and documentation in process industry: Hazardous area classification - basic concept - classification - selection of equipment based on hazardous area - Instrument protection - Concept - types - Fire and Gas (F & G) system - concept and block diagram (types of fire and gas detectors) - Safety Instrumentation System (SIS) - concept and block diagram - different layers - Instrumentation documents - Process Flow Diagrams (PFDs), P&I Diagram - Instrument loop diagrams - Graphical element - line symbol - Instrument identification from P&I Diagram - Examples - Heat exchanger and Distillation column Text / Reference: T/R Book title/Author R1 Curtis .D.Johnson, "Process control instrumentation technology", PHI R2 George stephanopoulos, "Chemical Process Control", PHI R3 Krishnakanth, "Computer based Industrial Control", PHI R4 John webb, "Programmable Logic Controllers", Macmillan Publishing R5 Liptak, "Handbook of Process Control" R6 Jovitha Jerome, "Virtual Instrumentation using LabVIEW", PHI Rajasekaran. S. Vijayalakshmi Pai. G.A. "Neural Networks, Fuzzy Logic and R7 Genetic Algorithms", Prentice Hall of India Private Limited, 2003 Timothy J.Ross, "Fuzzy logic with Engineering Applications", McGraw Hill, R8 1995 Alireza bahadori, "Hazardous area classification in petroleum and chemical plants R9 a guide to mitigating risk" Online Resources: Sl.No Website Link 1 <https://www.slideshare.net/debuddit/hazardous-area-classification-56437172> 2 <https://youtu.be/IBJnU1MJAts> ഹാസർഡസ് ഹേജർഡസ് definition/meaning ഹാസർഡസ് ഹേജർഡസ്. ഹാസർഡസ് ഹേജർഡസ് ഹാസർഡസ്, steps, application ഹാസർഡസ് ഹാസർഡസ് ഹാസർഡസ്. Technical terms English-ഹാസർഡസ് ഹാസർഡസ്.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle →
Components/steps →
Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure
→ Observation → Result →
Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION**Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING**Activities from the syllabus**

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

► **Define or explain Explain different types of process variables 3 Understanding Compare Feedback and Feed forward control. (3 marks)**

► **Define or explain 4 Understanding system Explain Cascade control system, Ratio control. (3 marks)**

► **Define or explain 4 Understanding system and Adaptive control system Explain Inferential control system, Split. (3 marks)**

► **Define or explain Range Control system, Override control 4 Understanding system and Auctioneering control system Basic Process control strategies -Single variable - Independent variable - Interactive variable - compound variable - multivariable control (with examples) - Feedback control system - Feed forward control system - Comparison - Cascade control system - Ratio control system - Adaptive control system -Inferential control system - Split Range Control system - Override control system-Auctioneering control system. Explain the use of Computers in Process Control and solve the. (3 marks)**

Module 2

► **Define or explain System, Supervisory control and Direct 4 Understanding Digital Control. (3 marks)**

► **Define or explain Describe Distributed control system 2**
Understanding Explain Programmable Logic Controller and. (3 marks)

► **Define or explain 3 Understanding its application. (3 marks)**

► **Define or explain Solve engineering problems using PLC 2 Applying**
Explain Supervisory Control And Data. (3 marks)

Module 3

► **Define or explain Explain Virtual instrumentation. (3 marks)**

► **Define or explain Understand LabVIEW and its applications. 2**
Understanding Understand the concept of Artificial Neural. (3 marks)

► **Define or explain 3 Understanding Network Compare the various**
Learning methods used. (3 marks)

► **Define or explain 2 Understanding in ANN. (3 marks)**

Module 4

► **Define or explain selection of equipments based on hazardous 3**
Understanding area Explain the concept of Instrument protection
Understanding. (3 marks)

► **Define or explain 3 and types. (3 marks)**

► **Define or explain Describe Fire and Gas system. (3 marks)**

► **Define or explain Explain safety instrumentation system 3**
Understanding Explain engineering documents used in. (3 marks)

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Explain different types of process variables**

Understanding Compare Feedback and Feed forward control.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **System, Supervisory control and Direct 4 Understanding Digital Control.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Explain Virtual instrumentation.**

Module 4

1-mark definition: State the meaning of **selection of equipments based on hazardous 3 Understanding area Explain the concept of**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Instrument protection Understanding.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link
- <https://www.slideshare.net/debuddit/hazardous-area-classification-56437172>
<https://youtu.be/IBJnU1MJAts>

IN-PAGE SEARCH**Search this lesson**

Prepared from the supplied official SITTTTR syllabus PDF for Course 5082. Verify institutional updates against the current official curriculum.